Saturation Throughput Analysis of Hybrid Access MAC Protocol in IEEE 802.11ax WLANs

Arthi S.
Prof. Neelesh B. Mehta.
Dept. of ECE, Indian Institute of Science Bangalore.

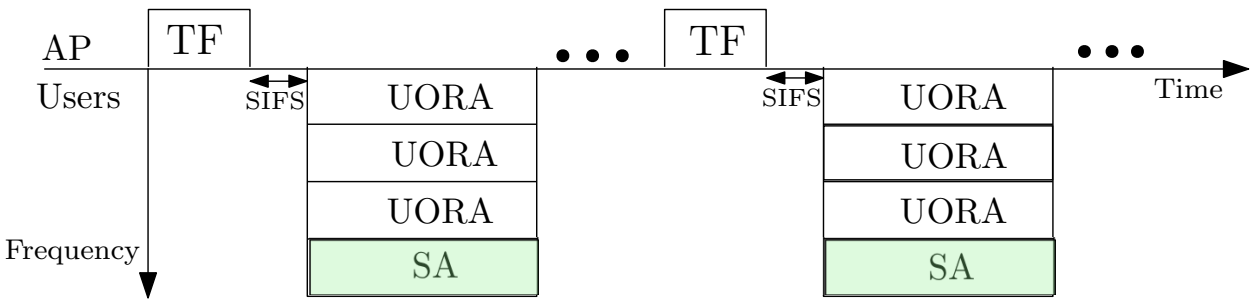
December 11, 2024.

Outline

- Hybrid access in 802.11ax WLANs
- Saturation throughput analysis
- Numerical results

Hybrid Access

- IEEE 802.11ax is the IEEE standard for Wi-Fi 6.
  - **Physical layer:**
    - Orthogonal frequency division multiple access (OFDMA).
    - Uplink multi-user multiple-input multiple-output (MU-MIMO).
  - **Medium access control (MAC) layer:**
    - OFDMA-based Hybrid access.
- **Hybrid access:**
  - UORA - uplink OFDMA random access, contention-based.
  - SA - scheduled access, contention-free.

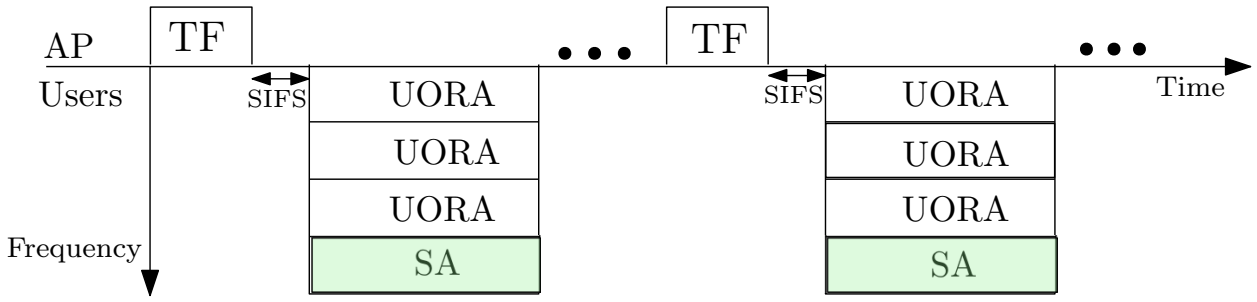


Contributions

- Analysis of IEEE 802.11ax with hybrid access.
  - UORA and SA saturation throughputs.
- **Characterize the variations in the number of users in UORA and SA.**
  - Differs from conventional assumptions [Bhattarai 2019], [Behara 2023].
- Incorporates impact of schedulers and discrete rate adaptation.
  - Random and channel-aware schedulers.

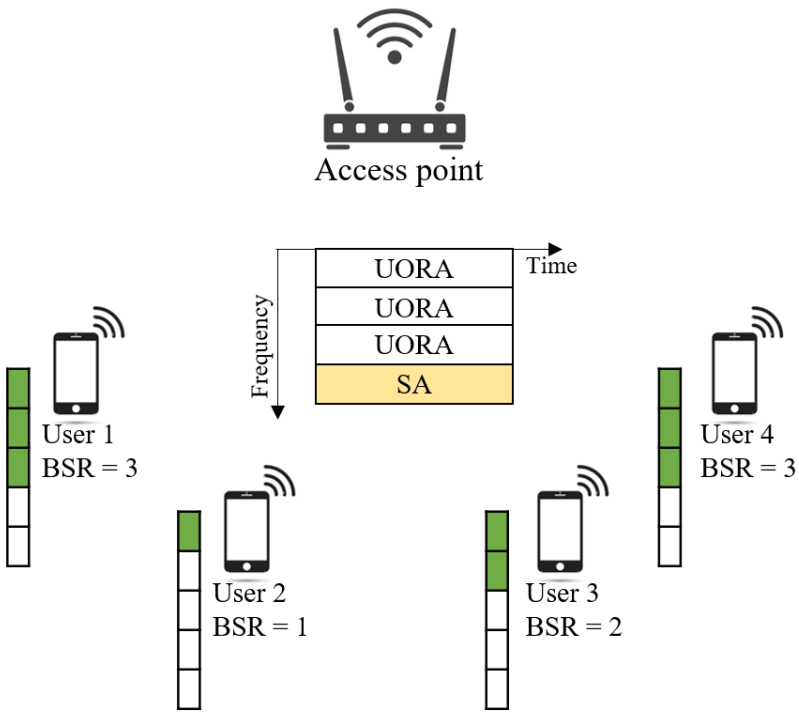
OFDMA

- Divides bandwidth into orthogonal resource units (RUs).
  - RUs are allocated to different users.



- AP manages OFDMA by transmitting trigger frames (TFs).
  - TF initiates OFDMA in uplink and downlink.
  - TF contains
    - RU allocation UORA and SA,
    - User-scheduling information for SA.
  - SIFS $\rightarrow$ Short inter-frame space.

Hybrid Access: UORA + SA



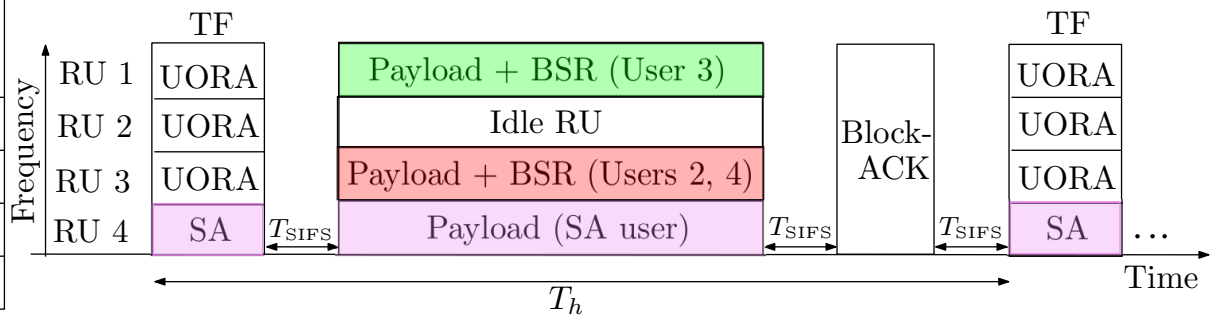
- AP allocates RUs for SA and UORA.
- **Scheduled access:**
  - AP schedules users on SA RUs.
  - Based buffer status reports (BSRs).
- **UORA:**
  - Users send BSRs to AP.
  - BSRs are sent with payload frames.

- SA is contention-free and improves throughput.
- UORA enables users to enter SA.

UORA: Back-off Process

- UORA follows exponential back-off-based contention mechanism.

User ID	Counter	
	Before TF	After TF
1	7	4
2	1	0
3	2	0
4	3	0



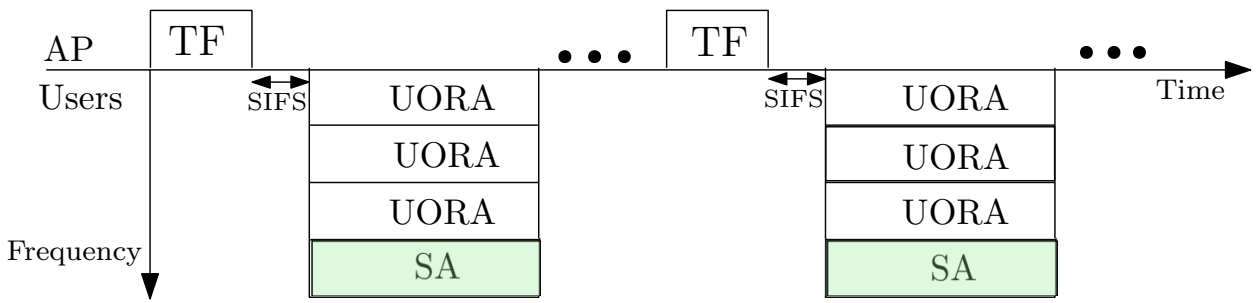
- Each UORA user uniformly samples a counter from $\{1, 2, \ldots, C\}$.
- A UORA user decrements counter if it receives TF with UORA RUs.
  - Attempts on a UORA RU if counter becomes zero.
  - **Success** $\rightarrow$ contention windows are reset to C_{min}.
  - **Collision** $\rightarrow$ contention window is doubled till C_{max}.

Differs from legacy random access in two ways:

- UORA back-off counter decrements are trigger-based.
- Back-off counter is decremented in bulk by N_{RA}.

Scheduled Access

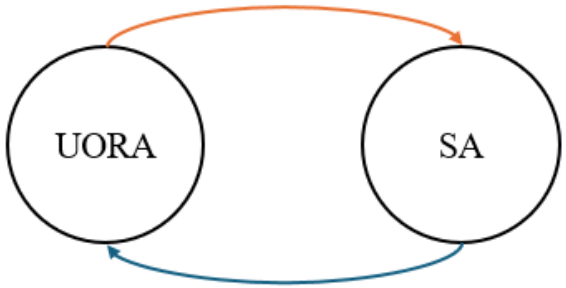
- User enters SA after transmitting BSR in UORA.
  - No longer participates in the UORA-based contention.
- Transmits its packets one by one in allocated RUs.
  - Scheduler determines user allocation in SA.
- User stays in SA for a random time $\rightarrow$ geometric random variable.

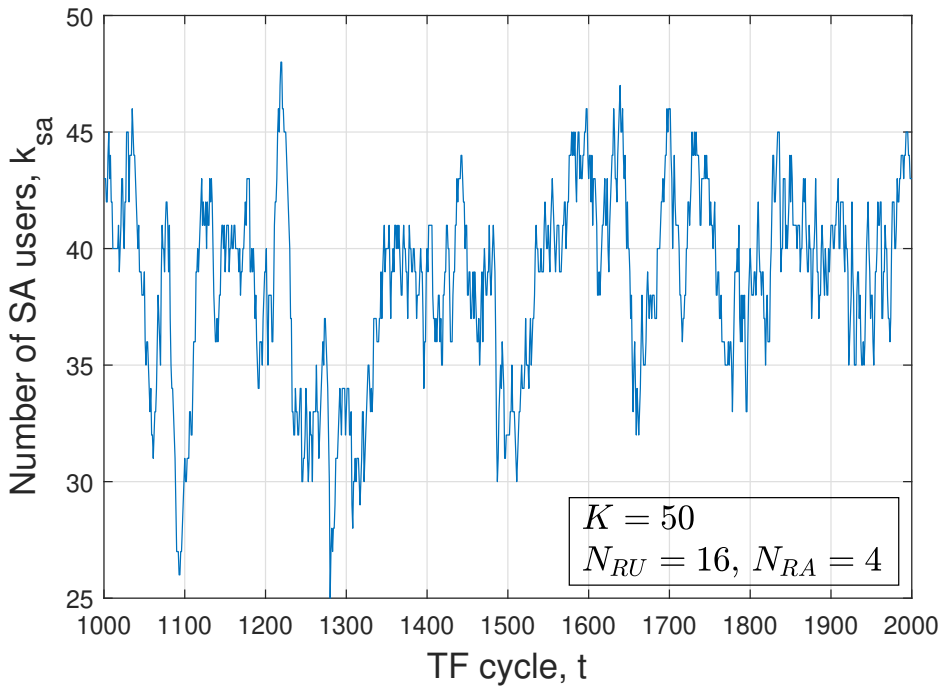


User returns to UORA after transmitting all its packets.

Dynamic Flow of UORA and SA Users

- **Users move between UORA and SA:**
  - Depends on RU allocation, MAC contention, and time user stays in SA.





Number of SA users is not a constant.

⇒ **Need to model.**

Saturation Throughput Analysis

Analysis: UORA Back-off Process

- Discrete-time Markov chain analysis:
  - Success and collision processes are coupled and correlated.
  - Number of states $= O(K^{\rho+1}) \rightarrow$ exact analysis is intractable.
    - K - Number of users and $\rho = \log_2(C_{\max}/C_{\min})$.

Decoupling Approximations [Bianchi 2000, Kumar 2007]:

- A user's attempt is independent of the states of all other users.
- At each attempt, collision occurs regardless of the number of re-transmissions.

- Back-off parameters for a given number of UORA users, k_{ra}:
  - Conditional collision probability, $\alpha(k_{\text{ra}})$
  - Attempt rate, $\beta(k_{\text{ra}})$

Analysis: UORA Back-off Process

- Discrete-time Markov chain analysis:
  - Success and collision processes are coupled and correlated.
  - Number of states $= O(K^{\rho+1}) \rightarrow$ exact analysis is intractable.
    - K - Number of users and $\rho = \log_2(C_{\max}/C_{\min})$.

Decoupling Approximations [Bianchi 2000, Kumar 2007]:

- A user's attempt is independent of the states of all other users.
- At each attempt, collision occurs regardless of the number of re-transmissions.

- Back-off parameters for a given number of UORA users, k_{ra}:
  - Conditional collision probability, $\alpha(k_{\text{ra}})$
  - Attempt rate, $\beta(k_{\text{ra}})$

Analysis: UORA Back-off Process

- Discrete-time Markov chain analysis:
  - Success and collision processes are coupled and correlated.
  - Number of states $= O(K^{\rho+1}) \rightarrow$ exact analysis is intractable.
    - K - Number of users and $\rho = \log_2(C_{\max}/C_{\min})$.

Decoupling Approximations [Bianchi 2000, Kumar 2007]:

- A user's attempt is independent of the states of all other users.
- At each attempt, collision occurs regardless of the number of re-transmissions.

- Back-off parameters for a given number of UORA users, k_{ra}:
  - Conditional collision probability, $\alpha(k_{\text{ra}})$
  - Attempt rate, $\beta(k_{\text{ra}})$

Analysis: UORA Back-off Process

- Discrete-time Markov chain analysis:
  - Success and collision processes are coupled and correlated.
  - Number of states $= O(K^{\rho+1}) \rightarrow$ exact analysis is intractable.
    - K - Number of users and $\rho = \log_2(C_{\max}/C_{\min})$.

Decoupling Approximations [Bianchi 2000, Kumar 2007]:

- A user's attempt is independent of the states of all other users.
- At each attempt, collision occurs regardless of the number of re-transmissions.

- Back-off parameters for a given number of UORA users, k_{ra}:
  - Conditional collision probability, $\alpha(k_{\text{ra}})$
  - Attempt rate, $\beta(k_{\text{ra}})$

Fixed-point Analysis

- **Conditional collision probability**

$$\alpha(k_{\text{ra}}) = 1 - \left(1 - \frac{\beta(k_{\text{ra}})}{N_{\text{RA}}}\right)^{k_{\text{ra}}-1}, \quad \text{for } 1 \leq k_{\text{ra}} \leq K.$$

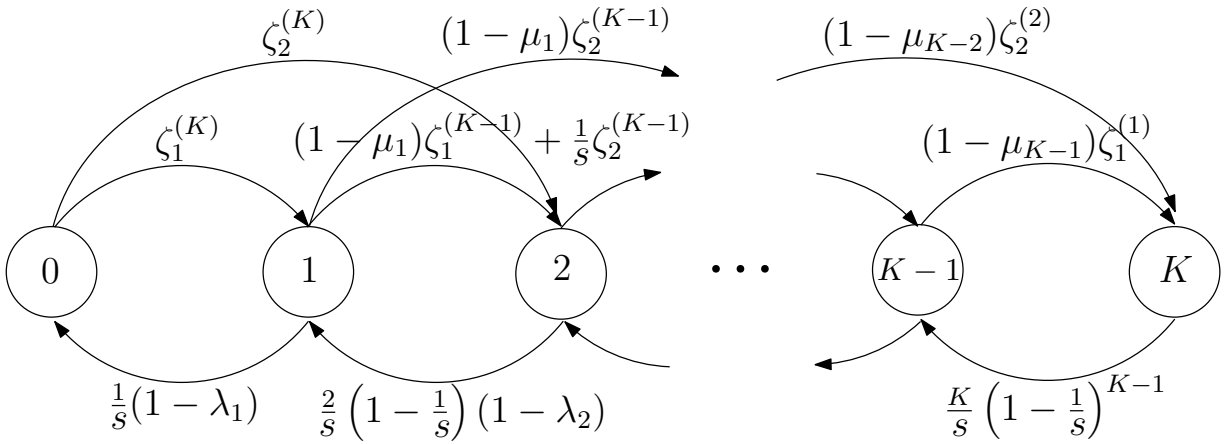
- **Attempt rate**

$$\beta(k_{\text{ra}}) = \frac{2N_{\text{RA}}(1 - 2\alpha(k_{\text{ra}}))}{(1 - 2\alpha(k_{\text{ra}}))(C_{\min} + N_{\text{RA}}) + \alpha(k_{\text{ra}})C_{\min}(1 - 2^{\rho}\alpha^{\rho}(k_{\text{ra}}))}.$$

- Expressed $\alpha(.)$ and $\beta(.)$ in terms of each other and system parameters.
- A unique fixed point can be found in $[0, 1]$.

Markov Model for Number of SA Users

- From decoupling approximations and geometric stay time in SA.
- **DTMC model for $N_{\mathsf{RA}} = 2$ and $N_{\mathsf{SA}} = 1$:**



State: Number of SA users in TF cycle t.

- $\zeta_n^{(k_{\mathsf{ra}})} = \Pr(n \text{ out of } k_{\mathsf{ra}} \text{ UORA users succeed})$.
- $\lambda_i = \Pr(\text{At least one UORA user succeeds})$.
- $\mu_i = \Pr(\text{At least one SA user leaves})$.

$\Rightarrow$ Written in terms of attempt rate and system parameters.

UORA Saturation Throughput

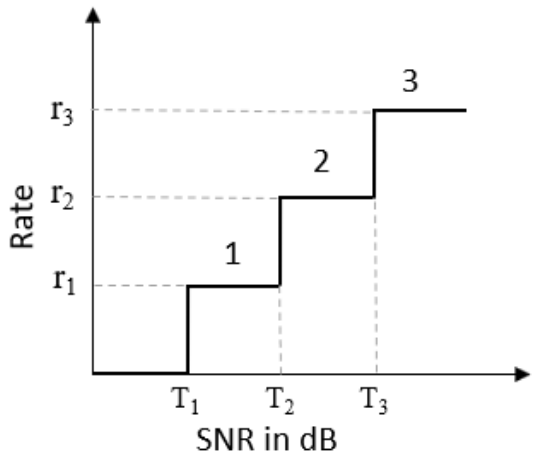
$$\Theta_{\text{UORA}} = \sum_{k_{\text{ra}}=1}^{K} \frac{T_P}{T_h} k_{\text{ra}} \nu(k_{\text{ra}}, N_{\text{RA}}) \bar{R} \Phi(k_{\text{ra}}).$$

- **Success probability of a UORA user:**

$$\nu(k_{\text{ra}}, N_{\text{RA}}) = \beta(k_{\text{ra}}) \left(1 - \frac{\beta(k_{\text{ra}})}{N_{\text{RA}}}\right)^{k_{\text{ra}}-1}.$$

- $\Phi(k_{\text{ra}})$ - Steady-state probability.
- T_P - payload, T_h - TF cycle duration.
- **Average throughput of a user:**

$$\bar{R} = \sum_m r_m \Pr(T_m \leq \text{SNR} < T_{m+1}),$$



Discrete rate adaptation

$$= \frac{2d_0^2}{\eta(d_{\max}^2 - 1)} \left(\frac{P_t K}{\sigma^2}\right)^{\frac{2}{\eta}} \sum_{m \in \Delta} \frac{r_m - r_{m-1}}{\Upsilon_m^{\frac{2}{\eta}}} \left[\gamma_{\text{inc}}\left(\frac{2}{\eta}, \frac{\Upsilon_m \sigma^2 d_{\max}^{\eta}}{P_t K d_0^{\eta}}\right) - \gamma_{\text{inc}}\left(\frac{2}{\eta}, \frac{\Upsilon_m \sigma^2}{P_t K d_0^{\eta}}\right)\right].$$

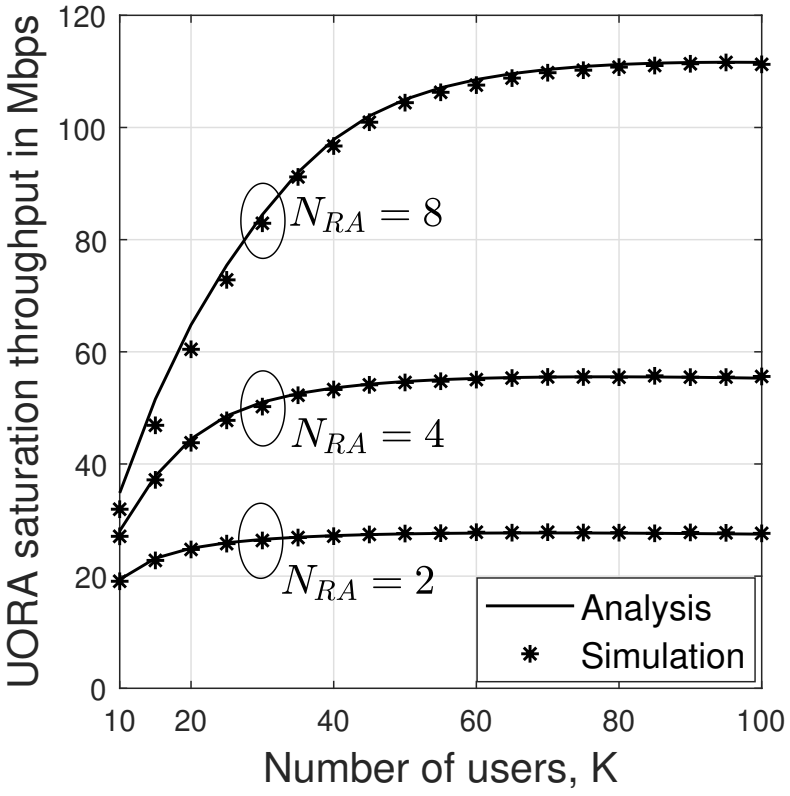
SA Saturation Throughput

$$\Theta_{\mathrm{SA}} = \sum_{k_{\mathrm{sa}}=1}^{K} \frac{\mathbb{E}\left[R(\mathcal{V})|k_{\mathrm{sa}}, N_{\mathrm{SA}}\right] T_P}{T_h} \Phi(k_{\mathrm{sa}}).$$

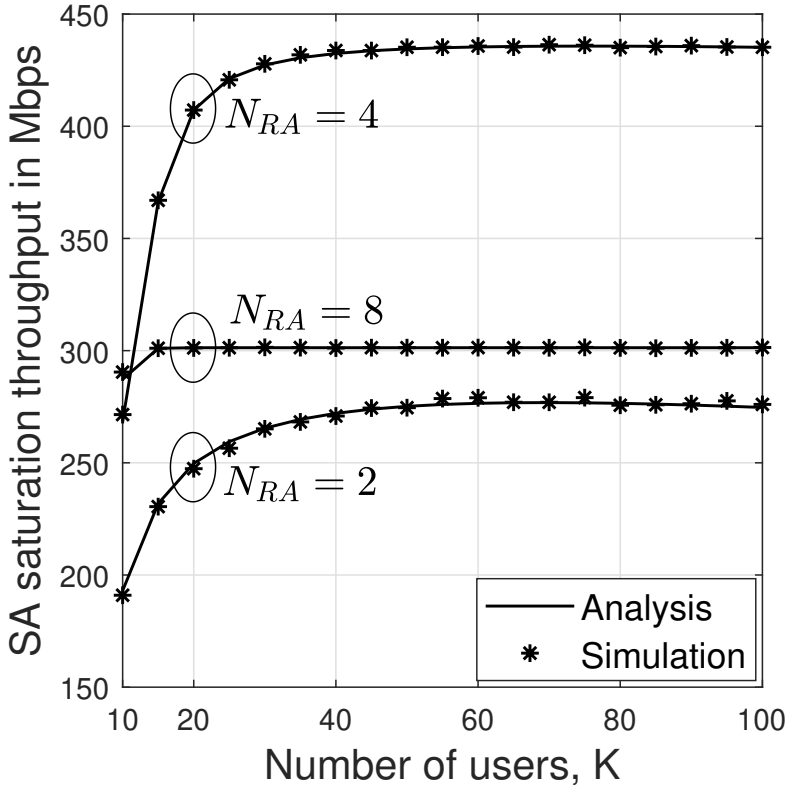
- $\mathbb{E}\left[R(\mathcal{V})|k_{\mathrm{sa}}, N_{\mathrm{SA}}\right]$ - average scheduler throughput
- **Random scheduler:**
  - Randomly allocates RUs to users.
  - $\mathbb{E}\left[R(\mathcal{V})|k_{\mathrm{sa}}, N_{\mathrm{SA}}\right] = \min\{k_{\mathrm{sa}}, N_{\mathrm{SA}}\}\bar{R}$.
- **Channel-aware scheduler:**
  - Maximizes sum throughput.
  - $\mathbb{E}\left[R(\mathcal{V})|k_{\mathrm{sa}}, N_{\mathrm{SA}}\right]$ is pre-computed and stored in a look-up table.

Numerical Results

UORA and SA Throughputs as a Function of K



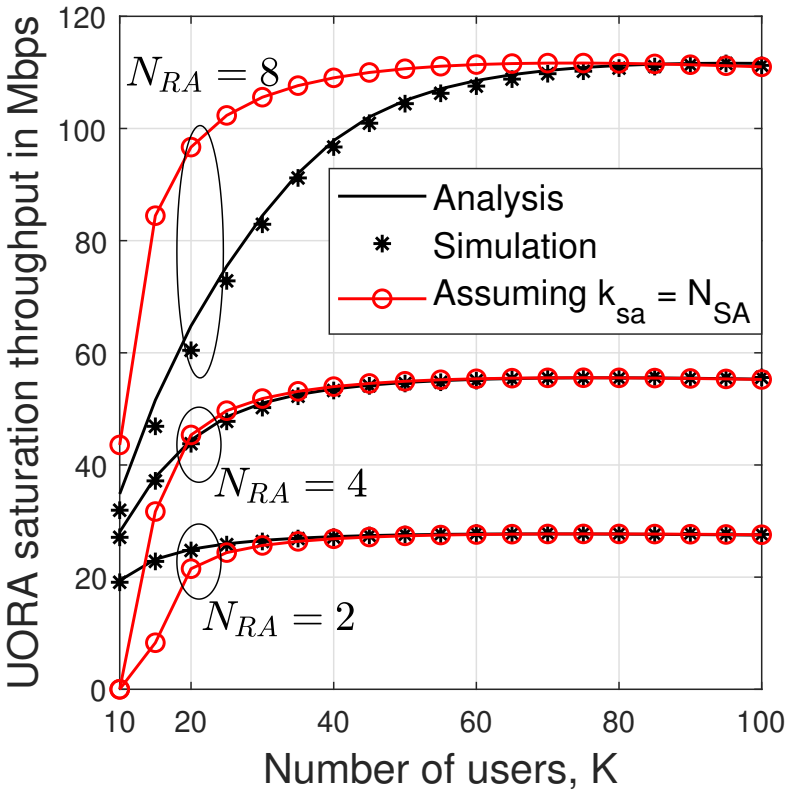
UORA saturation throughput



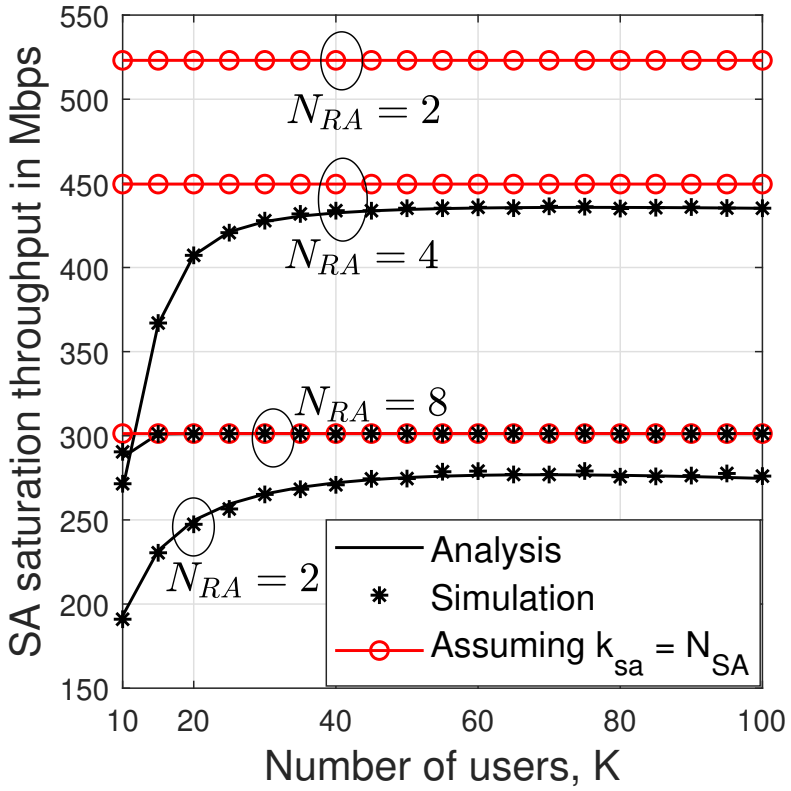
SA saturation throughput

- Bandwidth = 160 MHz and Total number of RUs, $N_{\mathrm{RU}} = 16$.
- UORA and SA throughputs increase as K increases.
- UORA throughput is smaller than the SA throughput.

UORA and SA Throughputs as a Function of K



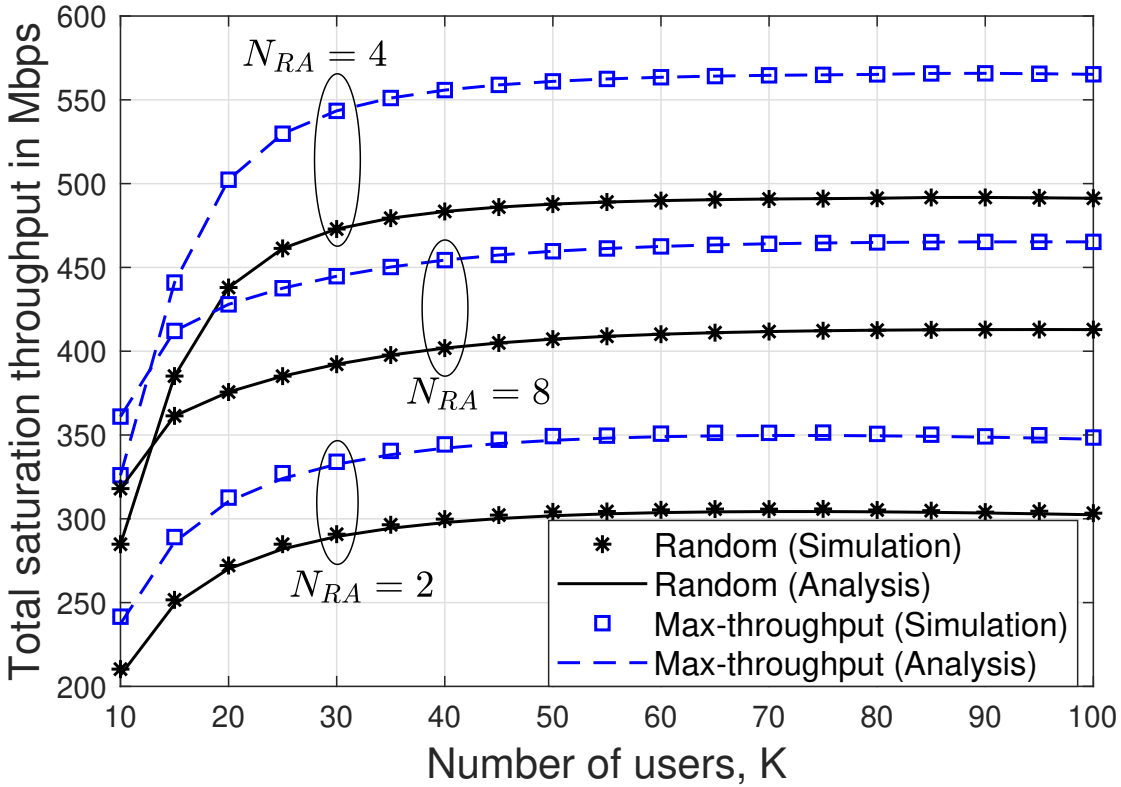
UORA saturation throughput



SA saturation throughput

- Bandwidth = 160 MHz and Total number of RUs, $N_{\mathrm{RU}} = 16$.
- UORA and SA throughputs increase as K increases.
- UORA throughput is smaller than the SA throughput.

Impact of Scheduler



- Max-throughput scheduler achieves higher throughput.
- Max-throughput scheduler's throughput depend on N_{RA}.

Conclusions

- Analyzed throughput of IEEE 802.11ax with UORA and SA.
  - Using decoupling approximations and Markov model.
  - Incorporated the flow of users between UORA and SA.
  - Evaluated the impact of schedulers on throughput.

Takeaways:

- UORA and SA throughputs depend on N_{RA}.
- Max-throughput scheduler's throughput depends on N_{RA}.
- Need to model variability in the number of SA and UORA users.

Thank you!

References

- S. Bhattarai, G. Naik, and J.-M. J. Park, "Uplink resource allocation in IEEE 802.11ax," in Proc. ICC, May 2019, pp. 1–6.
- A. Behara and T. G. Venkatesh, "Performance analysis and energy efficiency of MU-(OFDMA & MIMO) based hybrid MAC protocol of IEEE 802.11ax WLANs," IEEE Trans. Veh. Technol., vol. 72, no. 5, pp. 6474–490, May 2023.
- A. Kumar, E. Altman, D. Miorandi, and M. Goyal, "New insights from a fixed-point analysis of single-cell IEEE 802.11 WLANs," IEEE/ACM Trans. Netw., vol. 15, no. 3, pp. 588–601, Jun. 2007.
- G. Bianchi, "Performance analysis of the IEEE 802.11 distributed coordination function," IEEE J. Sel. Areas Commun., vol. 18, no. 3, pp. 535–547, Mar. 2000.